

Energy Optimization Model

The growing emphasis on clean energy transitions and the integration of renewable energy sources has catalysed the emergence of energy communities. These communities foster collective action to enhance energy efficiency, lower costs, and support local job creation. However, their integration into the energy system introduces challenges, particularly regarding the effective management of energy surpluses and deficits. Daily and seasonal energy storage systems, such as lithium and hydrogen technologies, play a pivotal role in addressing these imbalances.

Developing cost-efficient strategies for managing storage systems requires advanced modelling approaches, particularly those incorporating dynamic decision-making variables, diverse energy sources, and real-time energy demand patterns.

We now describe the problem which consists of minimizing the power exchanged with an external electricity supply network (grid). This can be interpreted as minimizing the total energy transferred to or drawn from the grid, reducing the costs associated with purchasing energy from the grid, or maximizing self-consumption to decrease dependency on the external network.

The solutions that we propose is described in the following.

Given as input the energy productions and the energy consumptions (expressed in Kw/h) of each time point of a particular day, we want to calculate the best quantity of energy to be taken from the battery with the aim to minimize to quantity of energy bought from the external grid and the quantity of energy feeded to the external grid. This means that the solution consists in maximizing the use of the battery and the charge of the battery from the on-site energy generation. Note that the quantity of production and consumption can be the results of a prediction made from the ML model.

Input

- The building name: for each specified building, some parameters like the maximum capacity of the storage, as well as the possible minimum and maximum battery level can be specified;
- Date: a date referred to a day;
- Unit: the unit of measurement, currently it is “kWh”;
- Initial Charge Percentage: the current percentage of the storage level;
- Productions and Consumptions for each time t;

Output

- Grid Transferred Power for each time t
- Storage Transferred Power for each time t
- Residual Storage power (at the end of the recommendation)

Exposed API from the service

URL	Input	Output	Note
/best_grid_transfer	<pre>{ "building": string, "date": string, "init_charge_percentage": float, "unit": string, "production": [{ "time": string, "value": float }, ...], }</pre>	<pre>{ "date": string, "unit": string, "discharge": [{ "time": string, "value": float }, ...], }</pre>	

	<pre> ...], "consumption": [{ "time": string "value": float } ... , time_execution_limit_secs": integer } </pre>	<pre> "charge": [{ "time": string, "value": float }, ...], "production": [{ "time": string, "value": float }, ...], "consumption": [{ "time": string, "value": 0 }, ...], "feed_in": [{ "time": string, "value": float }, ...], "from_grid": [{ "time": string, "value": float }, ...], final_charge_percentage: float } </pre>	
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